

Presentation

course “Essentials of computing systems”

LCC + LEFIS + LEI

Feb.2026

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course

- This course was conceived for **programmers and software engineers** who want to comprehend how the components of a computer work and how they affect the correctness and performance of their programs.
- Understanding how a computer works is fundamental for software engineers to comprehend the principles governing the execution of the programs they develop/maintain.
- It is also a fundamental knowledge if one wants to optimise the performance of a program, to write a compiler, or to develop an embedded system.
- **This course is focused on describing the fundamental aspects of computers, by studying their organisation and structure.**

the name of the game

- **System**: an identifiable and coherent set of components that interact to achieve a given objective.
- **Computing**: the use of computational systems to process information, solve problems, and perform tasks by following well-defined instructions (algorithms).

Computer Science

Informatics

Informatics Engineering

Information Systems

Artificial Intelligence.

topics

- Major pedagogical units:
 - Computer systems
 - Representation of information
 - Representation of numbers
 - IA32 instruction-set architecture
 - Cache memory
 - Code optimisations
- We assume that students know how to program a computer in a high-level language (e.g., C).
- Part of the contents of the course are addressed in Laboratórios Informática II (LEI).
- The contents of this course is continued in “Arquitetura de Computadores” (LEI; 2.º ano / 1.º semestre).

professors

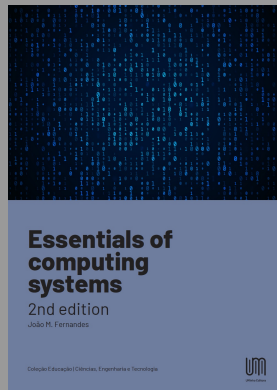
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António J. Esteves	AJE	LEI	PL4 PL6 PL7
João L. Sobral	JLS	LEI	PL5
Paulo R. Sousa	PRS	LEI	PL1 PL2 PL9 PL10



Professors can be reached through email, but they cannot guarantee immediate answer.

e-book

- A 100-page e-book was prepared to support the course.
- The e-book is free (published by UMinho-Editora).
- **Students must read the book.**
- Ambitious students must read other materials (e.g. suggested readings).
- Solutions to exercises are provided in the e-book.



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T-classes

- Slides in English / Speech in Portuguese.
- T-classes serve to expose/discuss course contents.
- Contents are totally described in the e-book.
- For LEI, there are two shifts, but we do not control the presences.

PL-classes

- PL/TP-classes follow TBL approach and serve to consolidate skills and knowledge.
- Students must prepare each PL/TP-class before entering the room.
- Some classes require access to a computer (gcc compiler or to `godbolt.org` website).
- Participation in 8 PL/TP-classes is mandatory. Relaxed to 5 (for 2025 students).
- Students may occasionally participate in a PL/TP class in which they are not enrolled.

schedule

semana			Teor. 1	Teor. 2	PL	
1	2026-02-02	2026-02-06	Apresent.	Ch1	sem aulas	
2	2026-02-09	2026-02-13	Ch1	Ch2	1.1 - 1.6	
3	2026-02-16	2026-02-20	Ch2	Ch3	2.1 - 2.10	
4	2026-02-23	2026-02-27	Ch3	Ch3	3.1 - 3.10	
5	2026-03-02	2026-03-06	Ch3	Ch3	3.11 - 3.19	
6	2026-03-09	2026-03-13	Ch4	Ch4	3.20 - 3.27	
7	2026-03-16	2026-03-20	Ch4	Ch4	3.28	
8	2026-03-23	2026-03-27	prep. teste	prep. teste	4.1 - 4.5	
9	2026-03-30	2026-04-03	não há aulas	não há aulas	não há aulas	
10	2026-04-06	2026-04-10	TESTE 1		revisão	TESTE 10.abril, 18:00
11	2026-04-13	2026-04-17	Ch5	Ch5	4.6 - 4.10	
12	2026-04-20	2026-04-24	Ch5	Ch6	5.1 - 5.4	
13	2026-04-27	2026-05-01	Ch6	talk IA	5.5 - 5.6	feriado sexta 01.05
14	2026-05-04	2026-05-08	prep. teste	não há aulas	6.1 - 6.4	última semana aulas
15	2026-05-11	2026-05-15	enterro Gata	enterro Gata	não há aulas	
16	2026-05-18	2026-05-22	TESTE 2		folga	TESTE 21.mai

assessment

- Final mark is given by: test #1 (50%) + test #2 (50%)
- A minimum of 9.5 points is needed to APPROVE.
- Participation in at least 8 (5) of the 11 TBL classes is a mandatory condition to APPROVE.
- Marks are rounded up
 11.00 → 11 11.33 → 12 11.67 → 12
- (2024) <10 52 10-12 113 13-15 76 16-18 16 19-20 1
 Success rate: 206/258 (80%)
- (2025) <10 141 10-12 44 13-15 34 16-18 10 19-20 7
 Success rate: 95/236 (40%)
- Use of AI during tests/exam is totally forbidden.
- Students may be required to explain their test responses orally.